

1. Gradient Descent Method in "Mathematics":

A. gradient descent method in one dimension:

$$x_{n+1} = x_n + \eta \cdot (-f'(x_n))$$

$y = f(x)$ is function
 x_{n+1} is the next point moving from x_n .

B. gradient descent method in multidimension:

$$\begin{bmatrix} x'_0 \\ x'_1 \\ \vdots \\ x'_n \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} + \eta \cdot \begin{pmatrix} -\frac{\partial f}{\partial x_0} \\ -\frac{\partial f}{\partial x_1} \\ \vdots \\ -\frac{\partial f}{\partial x_n} \end{pmatrix} \longrightarrow \vec{x}' = \vec{x} + \eta \cdot (-\nabla f)$$

$y = f(x_0, x_1, \dots, x_n)$ is function

$\begin{bmatrix} x'_0 \\ x'_1 \\ \vdots \\ x'_n \end{bmatrix}$ is the next point moving from $\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$

$\vec{x}' = \begin{bmatrix} x'_0 \\ x'_1 \\ \vdots \\ x'_n \end{bmatrix}$, is a vector.

$\vec{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$, is a vector.

C. step size, $|\eta \cdot (-f'(x_n))|$ or $\left| \eta \cdot \begin{pmatrix} -\frac{\partial f}{\partial x_0} \\ -\frac{\partial f}{\partial x_1} \\ \vdots \\ -\frac{\partial f}{\partial x_n} \end{pmatrix} \right|$, depends on learning rate (η) and (partial) derivative $\begin{pmatrix} -f'(x_n) \text{ or } -\frac{\partial f}{\partial x_0} \\ -\frac{\partial f}{\partial x_1} \\ \vdots \\ -\frac{\partial f}{\partial x_n} \end{pmatrix}$,

so step size will decrease as size of (partial) derivative decrease.

as gradient descent method converge:

$y = f(x)$ or $y = f(x_0, x_1, \dots, x_n)$ function converge at minimum point(s) $\longrightarrow$

the size of (partial) derivative approach 0 $\longrightarrow$

which implies that step size also approach 0.

D. if learning rate (η is always a positive number) is set to too large value,

$y = f(x)$ or $y = f(x_0, x_1, \dots, x_n)$ function converge at minimum point(s) $\longrightarrow$

the size of (partial) derivative approach 0 $\longrightarrow$

but learning rate is too big for step size to approach 0. $\longrightarrow$

gradient descent method tend to overshoot the solution and fail to converge.

2. Optimizing Problem in Perceptron Learning Algorithm:

in perceptron learning algorithm, we hope "predicted output P_y " & "output y (ground truth)" can be the same finally.

this is, we define various types of error functions as indicators to treat our optimizing problem,

whether we find out appropriate weights that minimize the value of error function.